

Intro to **Optimization under** **Uncertainty**

2026 Fall
AD7031



AI Applications



ChatGPT

Gemini



perplexity

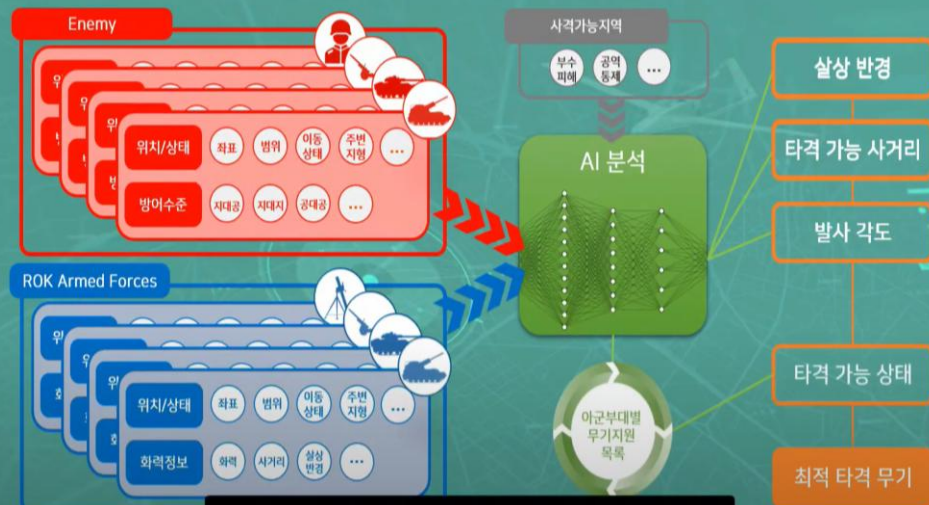


Claude

AI in Defense: Use Cases



METT-TC 기반 AI 분석



최적의 공격킬체인 구성 방안을 제안하게 됩니다



원료 또는 재료의 50% 넘게 사용하여 가공한 제품만 해당한다.

2. [검색된 정보를 바탕으로 생각해볼게요]

- 사용자는 설명절 기간에 허용되는 선물의 종류를 묻고 있습니다.
- 지식베이스에서 설명절 기간 동안 허용되는 선물의 가액 범위와 농수산물 선물의 종류에 대한 정보를 찾았습니다.
- 농수산물 및 농수산물가공품 선물은 30만원까지 허용되며, 농수산물에는 농산물, 수산물, 축산물, 임산물이 포함됩니다.

3. [최종 답변] 요청하신 질문은 설명절 기간에 허용되는 선물의 종류는 무엇인가요? 설명절 기간 동안 공직자에

GeDAI (Generative Defense AI)

국방 생성형 인공지능

ARTIFICIAL INTELLIGENCE

The way we train AI is fundamentally flawed

The process used to build most of the machine-learning models we use today can't tell if they will work in the real world or not—and that's a problem.

By Will Douglas Heaven

November 18, 2020

Why Robust AI?



Boston dynamic Atlas



Tesla Autopilot

Why Robust AI?



Boston dynamic Atlas



Tesla Autopilot

AI is not magic —
it is optimization under uncertainty



ChatGPT

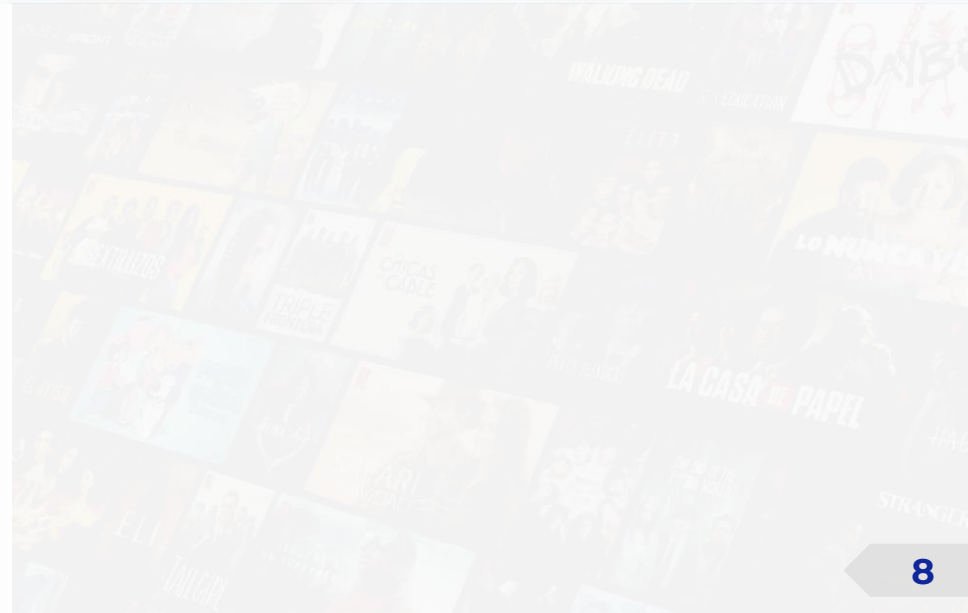
Gemini



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Optimization under Uncertainty: ML

Random vector $\xi = \begin{bmatrix} x \\ y \end{bmatrix}$ follows distribution $\mathbb{P}$ (i.e., $\xi \sim \mathbb{P}$), then the best model parameter θ^* is

$$\theta^* \in \arg \min_{\theta} \mathbb{E}_{\mathbb{P}} [\ell(y, f_{\theta}(x))]$$

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Many **loss (objective) functions** ℓ are used in ML

Regression :

$$\mathbf{MSE}(\cdot) = \|y - f_{\theta}(x)\|_2^2$$

Classification :

$$\mathbf{0-1 loss}(\cdot) = \begin{cases} 1, & \text{if } \text{sign}(f_{\theta}(x)) \neq \text{sign}(y) \\ 0, & \text{if } \text{sign}(f_{\theta}(x)) = \text{sign}(y) \end{cases}$$

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$\Rightarrow$ we can formulate 0-1 loss as IP!

Optimization under Uncertainty: ML

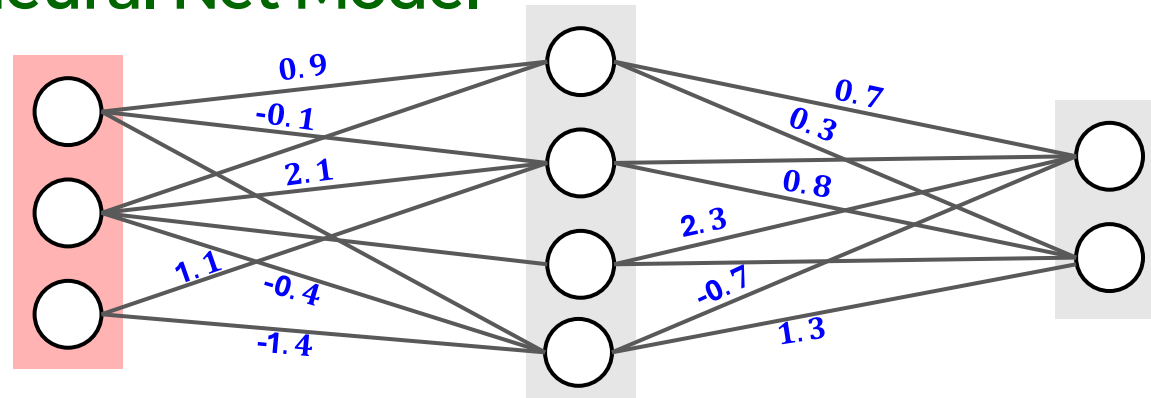
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Linear Model

$$f_{\theta}(\mathbf{x}) = \theta^{\top} \mathbf{x} = \sum_i x_i \theta_i$$

Neural Net Model



Input

Hidden

Output

Optimization under Uncertainty: ML

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ML $\subset$ optimization under uncertainty

➤ OR :

$$\begin{aligned} \min_x \quad & \mathbb{E}_{\mathbb{P}} [f(x, \xi)] \\ \text{s.t.} \quad & \text{Prob}[g(x, \xi) \leq 0] \geq \alpha \end{aligned}$$

➤ Control

$$\begin{aligned} \min_{u_1, u_2, \dots, u_{T-1}} \quad & \mathbb{E}_{\mathbb{P}} [\sum_{t=1}^{T-1} c_t(\tilde{x}_t, u_t, \xi_t) + c_T(\tilde{x}_T)] \\ \text{s.t.} \quad & \tilde{x}_{t+1} = f_t(\tilde{x}_t, u_t, \xi_t) \quad \forall t = 1, \dots, T-1 \end{aligned}$$

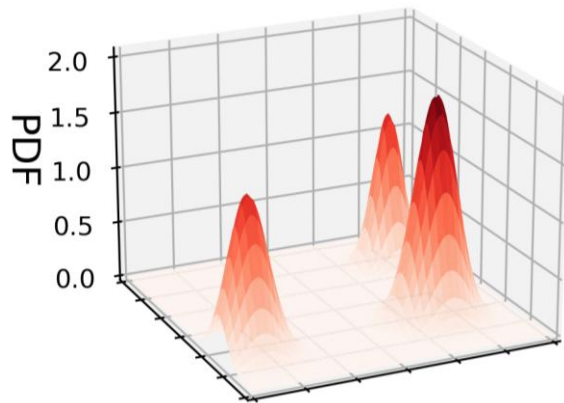
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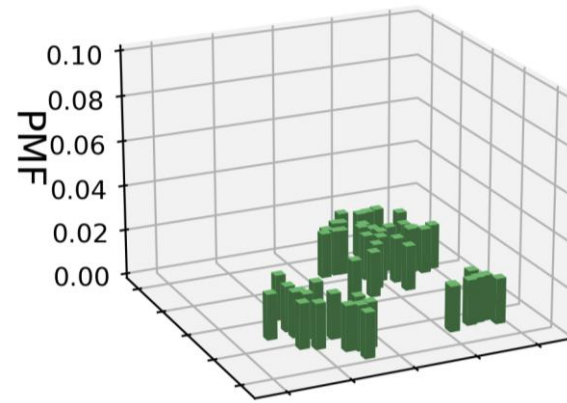
$$\theta^* \in \arg \min_{\theta} \mathbb{E}_{\mathbb{P}} [\ell(y, f_{\theta}(x))]$$

$\mathbb{P}$ unknown $\Rightarrow \theta^*$ not available

Only N samples $= \{\xi_1, \xi_2, \dots, \xi_N\}$ available: $(\xi_i = \begin{bmatrix} x_i \\ y_i \end{bmatrix})$



True distribution $\mathbb{P}$



Empirical distribution $\hat{\mathbb{P}}_N$

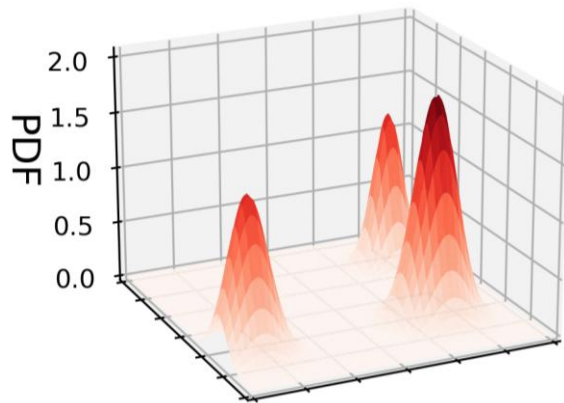
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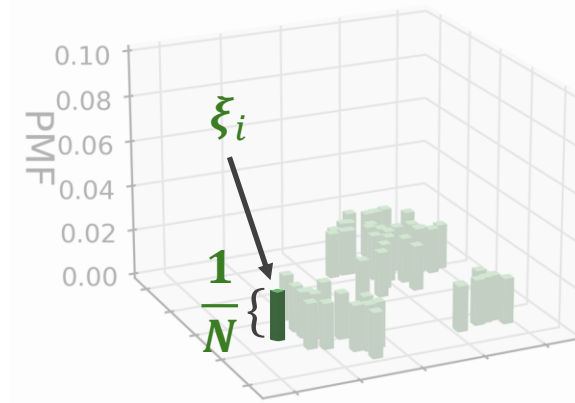
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Data-Driven Optimization

Assume $\xi \sim \hat{\mathbb{P}}_N$, then the best empirical parameter $\hat{\theta}_N$ is

$$\begin{aligned} \hat{\theta}_N &\in \arg \min_{\theta} \mathbb{E}_{\hat{\mathbb{P}}_N} [\ell(y, f_{\theta}(x))] \\ &= \arg \min_{\theta} \frac{1}{N} \sum_{i=1}^N \ell(y_i, f_{\theta}(x_i)) \end{aligned}$$

Optimization under Uncertainty: ML

True Problem

$$\theta^* \in \arg \min_{\theta} \mathbb{E}_{\mathbb{P}} [\ell(\mathbf{y}, f_{\theta}(\mathbf{x}))]$$

$\neq$

$$\hat{\theta}_N \in \arg \min_{\theta} \mathbb{E}_{\hat{\mathbb{P}}_N} [\ell(\mathbf{y}, f_{\theta}(\mathbf{x}))]$$

Data-Driven Problem

Optimization under Uncertainty: ML

True Problem

$$\theta^* \in \arg \min_{\theta} \mathbb{E}_{\mathbb{P}} [\ell(\mathbf{y}, f_{\theta}(\mathbf{x}))]$$

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Data-Driven Problem

If $\hat{\mathbb{P}}_N \approx \mathbb{P}$, then $\hat{\theta}_N$ is a good approximation to θ^*

- In ML, $\hat{\theta}_N$ is called Empirical Risk Minimizer (ERM)
- In OR, it is called Sample Average Approximation (SAA)

Optimization under Uncertainty: ML

True Problem

$$\theta^* \in \arg \min_{\theta} \mathbb{E}_{\mathbb{P}} [\ell(\mathbf{y}, f_{\theta}(\mathbf{x}))]$$

$\neq$

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Data-Driven Problem

Optimization under Uncertainty: ML

$$\theta^* \neq \hat{\theta}_N !$$

If $\hat{\mathbb{P}}_N$ poorly approximates $\mathbb{P}$ —especially with sparse data—
 $\hat{\theta}_N$ may perform poorly.

Optimization under Uncertainty: ML

True Problem

$$\theta^* \in \arg \min_{\theta} \mathbb{E}_{\mathbb{P}} [\ell(\mathbf{y}, f_{\theta}(\mathbf{x}))]$$

$\neq$

$$\hat{\theta}_N \in \arg \min_{\theta} \mathbb{E}_{\hat{\mathbb{P}}_N} [\ell(\mathbf{y}, f_{\theta}(\mathbf{x}))]$$

Data-Driven Problem

If $\hat{\mathbb{P}}_N$ poorly approximates $\mathbb{P}$ —especially with sparse data—

$\hat{\theta}_N$ may perform poorly.

Distributionally Robust Optimization (DRO)

Idea : Find the best $\hat{\theta}_N^{DRO}$, expecting the worst distribution :

$$\min_{\theta} \mathbb{E}_{\hat{\mathbb{P}}_N} [\ell(\mathbf{y}, f_{\theta}(\mathbf{x}))]$$

Distributionally Robust Optimization (DRO)

Idea : Find the best $\hat{\theta}_N^{DRO}$, expecting the worst distribution :

$$\min_{\theta} \max_{Q \in \mathcal{P}_{\lambda}(\hat{\mathbb{P}}_N)} \mathbb{E} [\ell(\mathbf{y}, f_{\theta}(\mathbf{x}))]$$

Distributionally Robust Optimization (DRO)

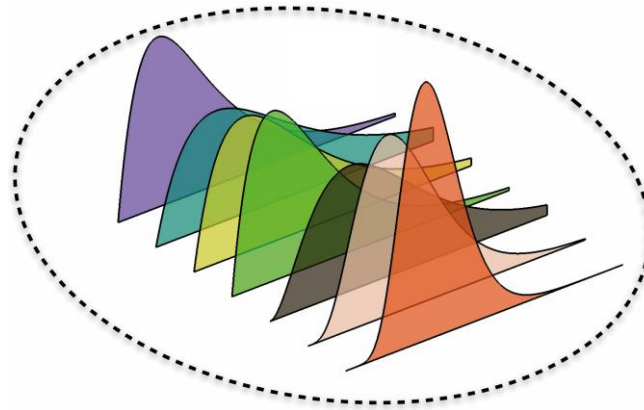
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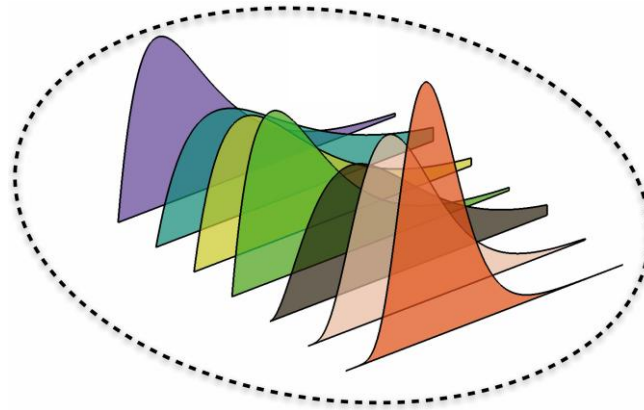


$\mathcal{P}_{\lambda}(\cdot)$: Ambiguity set

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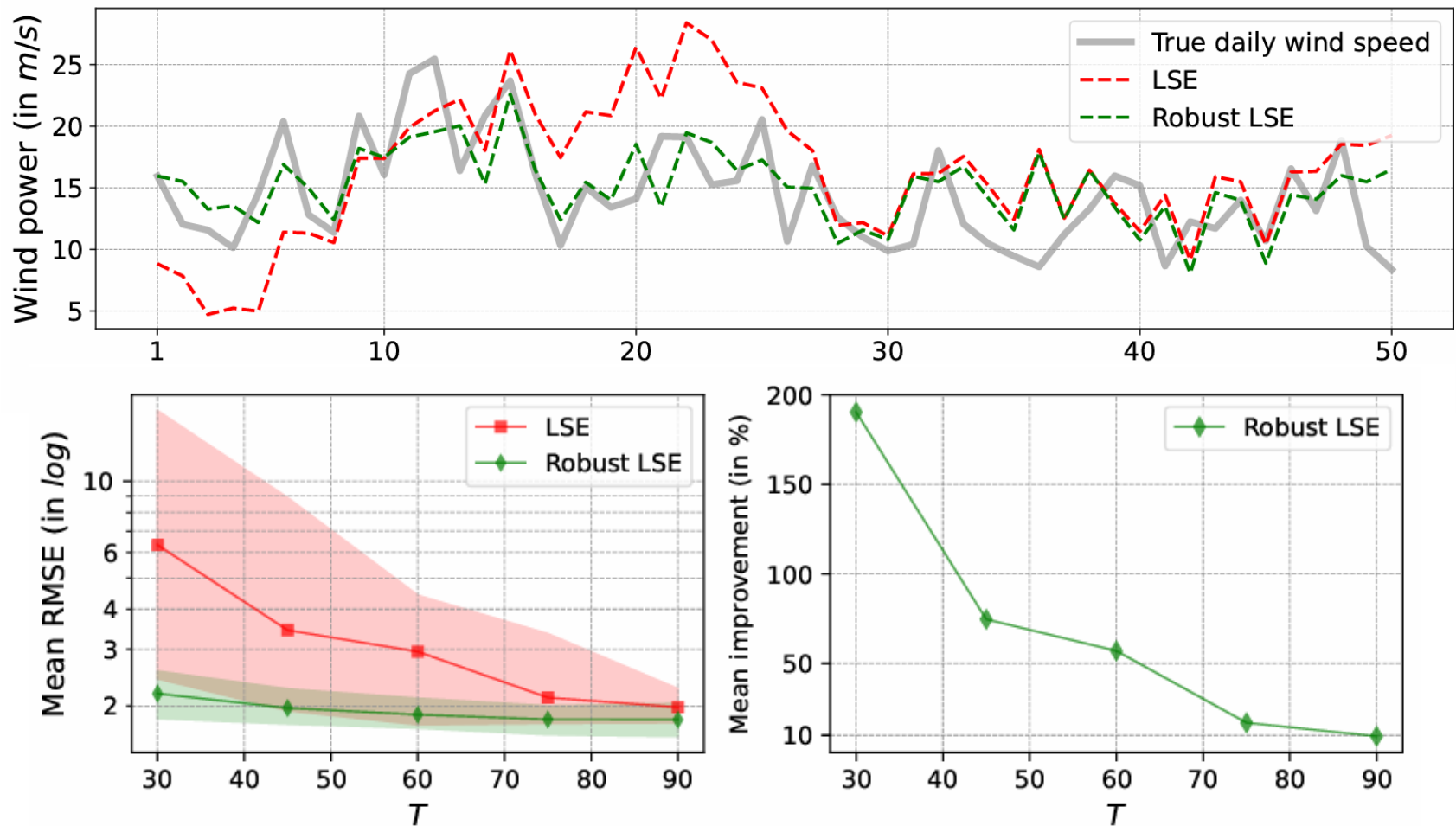


$\mathcal{P}_{\lambda}(\cdot)$: Ambiguity set

$\hat{\theta}_N^{DRO} \Rightarrow$ Robust performance under data scarcity

Machine Learning

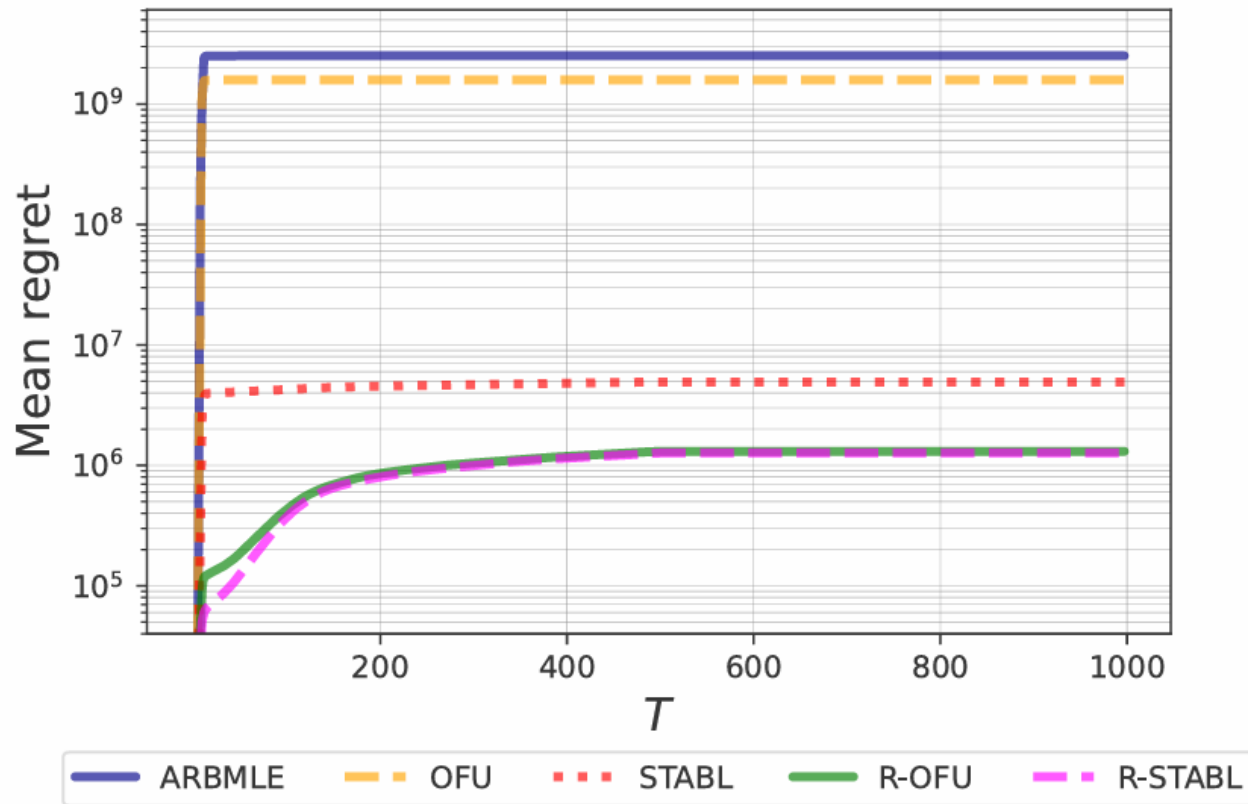
DRO-based NN model: Short-term Wind prediction ⁴⁾



⁴⁾ Park et al., "Robust System Identification", *ICLR*, 2025

Machine Learning

DRO-based Reinforcement Learning : Boeing 747 control ⁴⁾



⁴⁾ Park et al., "Robust System Identification", *ICLR*, 2025

Operations Research

DRO SDDP (Stochastic Dual Dynamic Programming) : Portfolio Optimization ⁶⁾

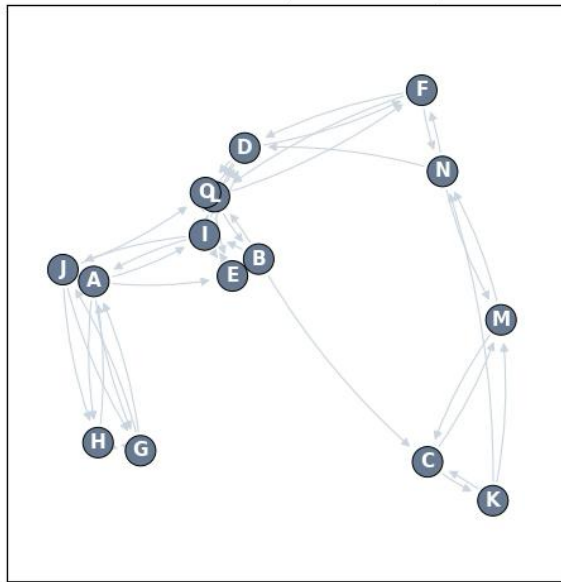
Dataset	Scheme	Utility	Mean return	Std. Dev.	Sharpe ratio
10 Industry Fama-French	DDR-SDDP	3.19E-02	3.41E-02	6.35E-02	4.77E-01
	DD-SDDP	2.65E-02	2.90E-02	6.95E-02	3.62E-01
	Independent	1.58E-02	1.76E-02	6.23E-02	2.21E-01
	HMM-SDDP	2.43E-02	2.54E-02	4.70E-02	4.59E-01
	Equal	-2.15E-03	-1.97E-03	2.74E-02	-3.52E-02
12 Industry Fama-French	DDR-SDDP	2.49E-02	2.67E-02	5.97E-02	3.82E-01
	DD-SDDP	2.35E-02	2.59E-02	6.90E-02	3.20E-01
	Independent	2.01E-02	2.20E-02	6.32E-02	2.88E-01
	HMM-SDDP	1.97E-02	2.09E-02	4.97E-02	3.42E-01
	Equal	-1.52E-03	-1.36E-03	2.72E-02	-3.17E-02
12 Industry Fama-French	DDR-SDDP	1.38E-02	1.43E-02	3.61E-02	2.89E-01
	DD-SDDP	7.34E-03	9.45E-03	6.86E-02	8.16E-02
	Independent	3.43E-03	5.34E-03	6.51E-02	2.28E-02
	HMM-SDDP	5.68E-03	6.09E-03	3.43E-02	6.53E-02
	Equal	-5.00E-02	-4.87E-02	2.73E-02	-3.18E-01

6) Park et al., "Data-driven Stochastic Dual Dynamic Programming", *SIOPT*, 2026

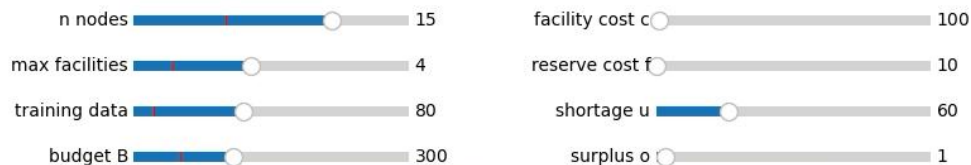
Operations Research

DRO Two-stage LAP (Location-Allocation Problem) : Drone scheduling ⁶⁾

Candidate network & transport costs — press Solve



● candidate site



Two-stage LAP (Sample Average Approximation)

Stage 1 (here-and-now, before the disaster)

$$\min_{\mathbf{x}, \mathbf{r}} \sum_j c_j x_j + \sum_j f_j r_j + \frac{1}{N} \sum_{n=1}^N Q(\mathbf{x}, \mathbf{r}; \hat{\xi}^n)$$

$$\text{s.t. } \sum_{j \in \mathcal{N}'} r_j \leq B, \quad \sum_{j \in \mathcal{N}'} x_j \leq p$$

$$r_j \leq B x_j, x_j \in \{0, 1\}, r_j \geq 0 \quad \forall j \in \mathcal{N}'$$

Stage 2 (recourse for scenario n, after demand is seen)

$$Q(\mathbf{x}, \mathbf{r}; \hat{\xi}^n) = \min_{\mathbf{y}, \mathbf{q}, \mathbf{s}} \sum_a d_a y_a + \sum_j u_j q_j + \sum_j o_j s_j$$

$$\text{s.t. } (in_j - out_j) + q_j - s_j = \hat{\xi}_j^n - r_j \quad \forall j \in \mathcal{N}'$$

$$0 \leq y_a \leq \bar{y} \quad \forall a \in \mathcal{A}; \quad 0 \leq q_j \leq \hat{\xi}_j^n, s_j \geq 0$$

DECISION VARIABLES (what we choose)

s1: $\mathbf{x}$ open facilities, $\mathbf{r}$ reserves; s2: $\mathbf{y}$ flows, $\mathbf{q}$ shortages, $\mathbf{s}$ surpluses

UNCERTAINTY (seen only after stage 1)

$\hat{\xi}^n$ = demand vector in scenario n ($\hat{\xi}_j^n$ at node j)

PARAMETERS (given numbers)

c_j open, f_j reserve, B budget, p cap; d_a transport, u_j short, o_j surplus, N #scen

SETS & INDICES

graph $G = (\mathcal{N}', \mathcal{A})$: $\mathcal{N}'$ nodes, $\mathcal{A}$ arcs; $j \in \mathcal{N}'$, $a \in \mathcal{A}$, $n = 1, \dots, N$



press Solve to choose where to build
and how much to stock

trials: 300

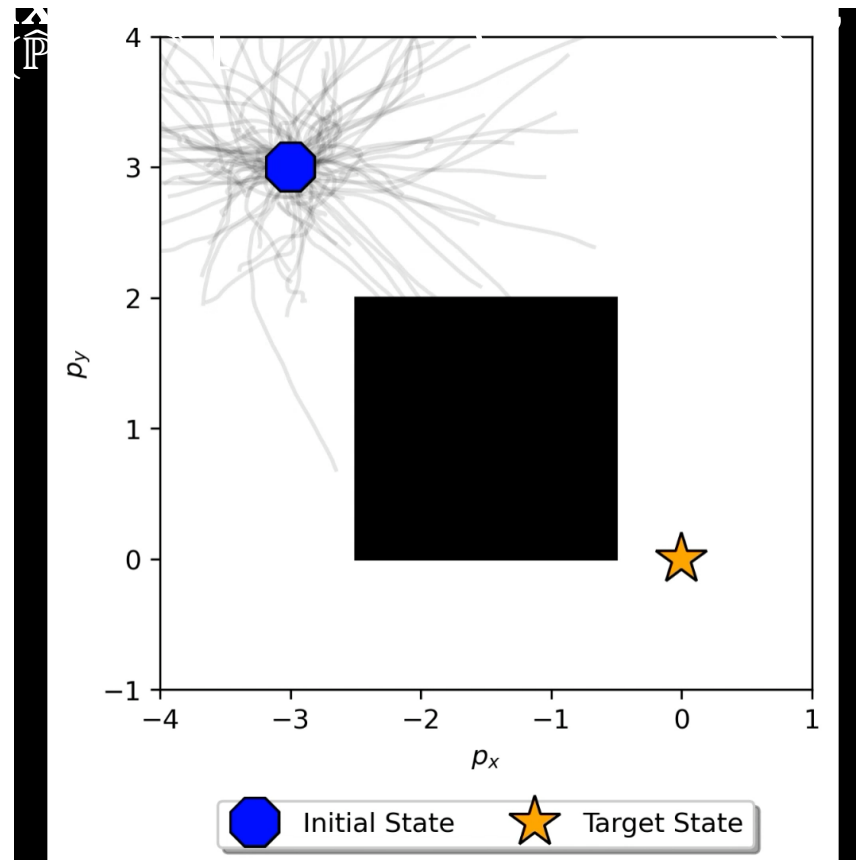
band: IQR 25-75

Run trials

Cost vs N curve

Control

DRO PIC(Path Integral Control) : Robotics simulation ⁷⁾

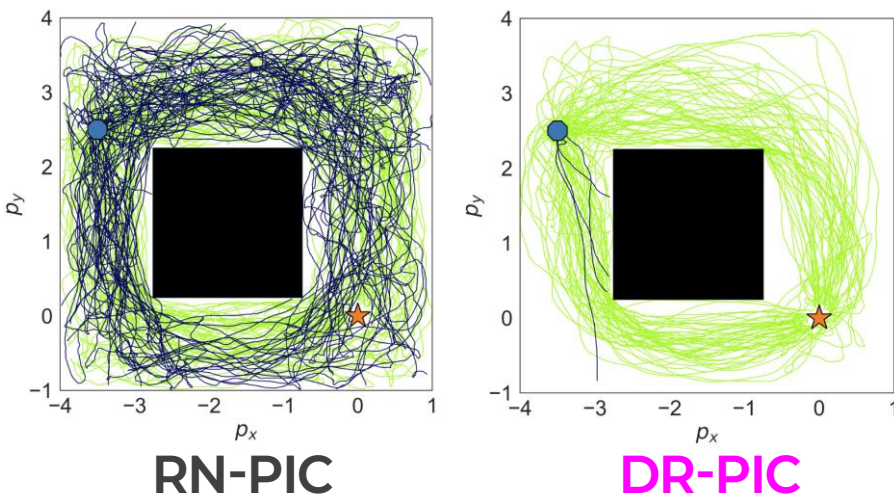


⁷⁾ Park et al., "Distributionally Robust Path Integral Control," *Proc. ACC*, 2025.

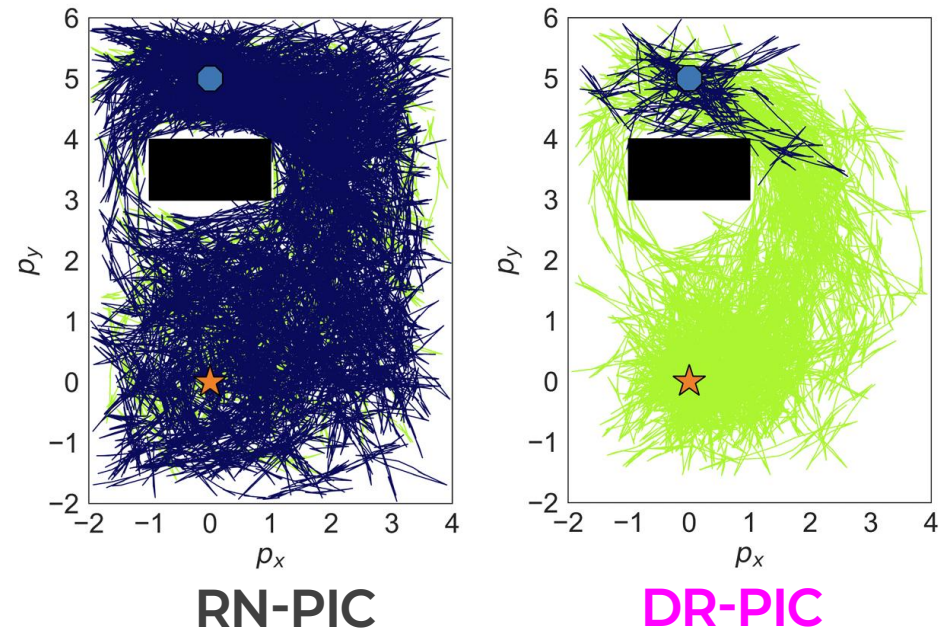
Control

DRO PIC(Path Integral Control) : Robotics simulation ⁷⁾

➤ 2D Particle robot



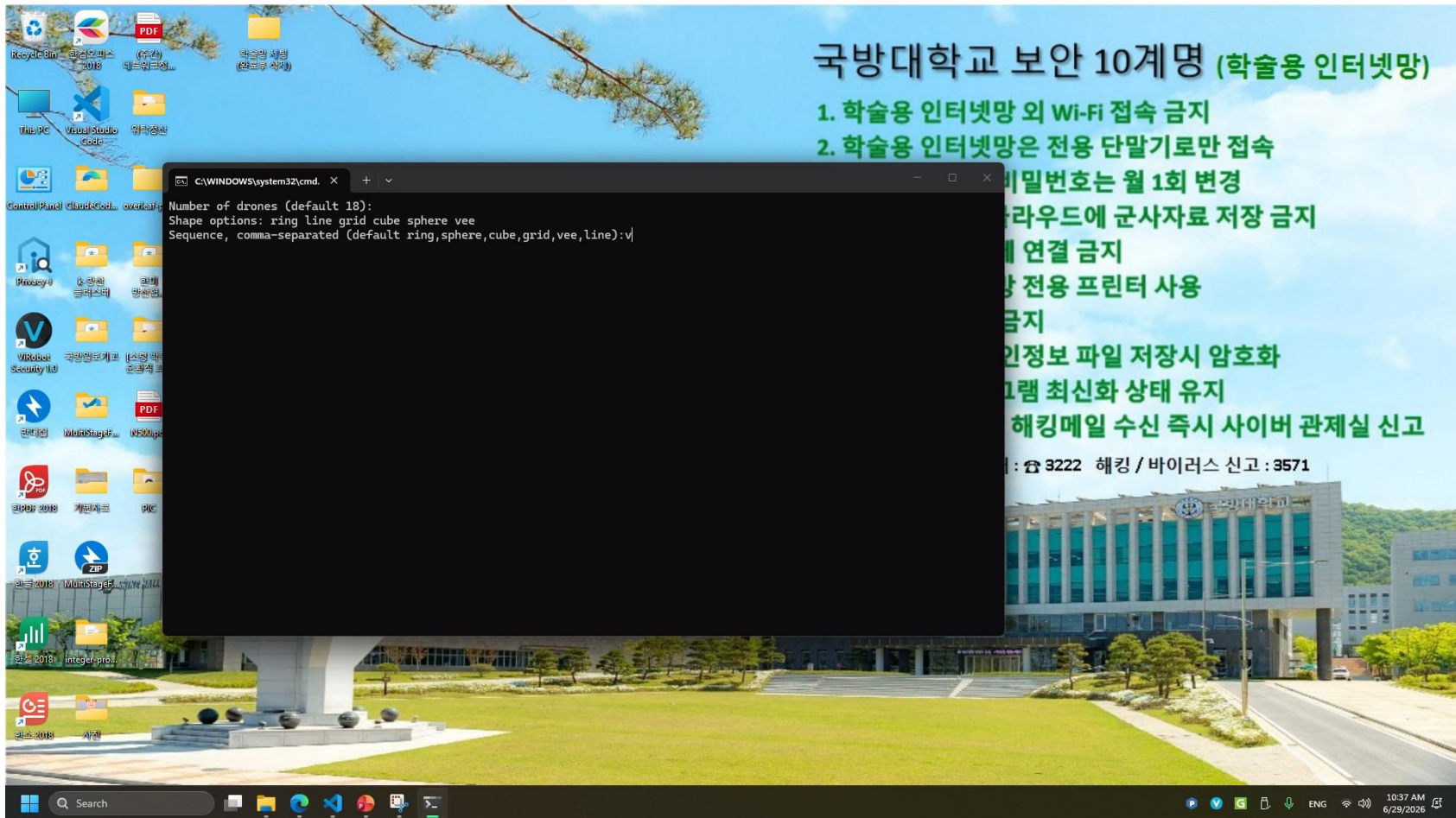
➤ Unicycle robot



⁷⁾ Park et al., "Distributionally Robust Path Integral Control," *Proc. ACC*, 2025.

Control

DRO PIC(Path Integral Control) : Robotics simulation ⁷⁾



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Thanks!
Any question?

